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TomMac Report

Group 11

Introduction

TomMac must manage its resources efficiently year-round with changing tomato availability, prices, and specific product demands. We aim to build a model that TomMac can use to determine the optimal quantity of tomatoes to purchase, what mix of products to produce, and the inventory levels to maintain, while also keeping business constraints in consideration. We will first produce a base case corresponding to the current situation of TomMac, suggest a future scenario based on current predictions and end with additional considerations for the company.

Stage 1: Base Case

The total income and cost for the year are £136,498.0 and £112,006.3, respectively. The maximised **total profit is £24,491.7**. All the optimised values for the base case can be referenced in the appendix (tables 1–10, 13–20). All values in this report are quoted to an accuracy of 1 decimal digit.

As can be seen in the profit breakdown table (table 1 in appendix), there is negative profit in Jul-Sep, indicating the increased purchase of stock to meet contract and demand needs for upcoming seasons. When adding a requirement for non-negative quarterly profit to the model, keeping everything else the same, we arrive at the figures from table 2 (see appendix). Annual profit drops to slightly below £24,000 and Q3 profit is exactly 0.

Going further, this report will keep reporting results for the first case, i.e. when negative quarterly profit is allowed for the sake of higher annual profit, but we leave the choice of choosing which model to use at the discretion of TomMac.

Looking at the income breakdown table (table 3 in appendix), we can see that the income is highest during the second and third quarters. Additionally, the cost of making products, stock, and total cost are highest in the third quarter (table 4). This is due to the cheaper cost of tomatoes during the spring and summer months. Thus, production is higher during these quarters to take advantage of higher profit margin potential.

The quantity of each variety of tomato is quoted in kilos by quarters in table 5. Yearly costs for Beef, Plum, and Moneymaker tomatoes are £5,969.3, £16,676.2, and £14,800 respectively. Comparing the data in table 5 with TomMac's tomato supply limits, we know that all Beef tomatoes are used in the 1st and 4th quarter. All Moneymaker tomatoes are used and reach the supply limit in each quarter. Although Plum tomatoes are used in each quarter, they do not reach supply limit.

Moneymaker tomatoes are the cheapest variety of tomatoes, and are in the middle in terms of their solid-liquid ratio, which explains their usefulness for all products and why they reach the supply limit. As for the Plum tomatoes, they have the highest solid-liquid ratio and are therefore essential for more solid products, Paste and Canned. However, they are the most expensive tomato variety, which is why they never reach the supply limit.

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The number of products in stock (quoted in kg) is shown in the figure below, detailed information along with costs can be found in table 6 in the appendix.

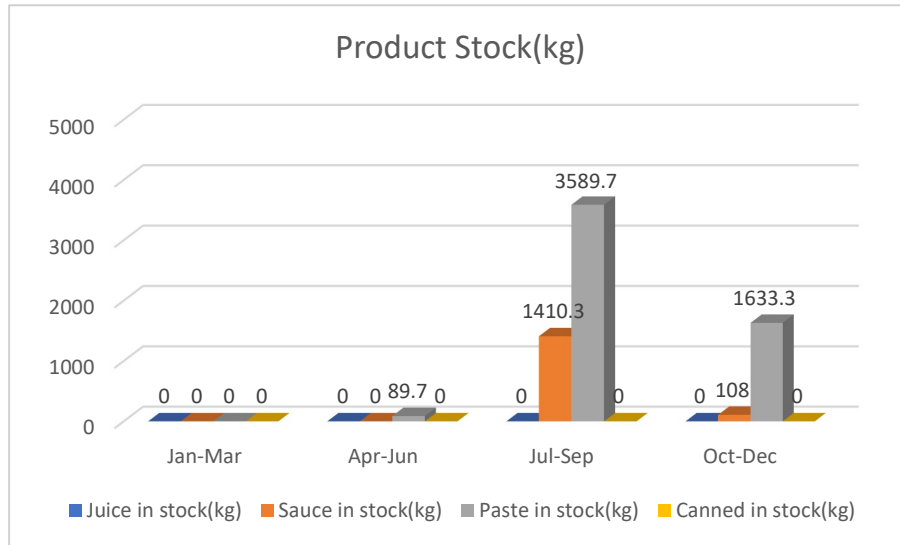


Figure 1. Amount of products stored per quarter.

From the stock information we observe the following:

- The warehouse is full at the end of 3rd quarter.
- Sales of Canned tomatoes equal demand in all quarters.
- The Juice and Canned are the most popular products and are out of stock of all times.
- The Paste is the most frequent product in stock.

Juice and Canned are always out of stock for all quarters. Since their reduction factor is 1, we need the least amount of tomatoes to produce them, so the profit from selling them depends slightly less on seasonal tomato cost. Paste, in turn, has the lowest reduction factor (0.5), meaning its profit will depend the most on tomato prices. That is why it is made in bulk when resources are cheapest. Its reduction factor also explains why less units are made than Canned, which will be covered in the next section.

Factors limiting profit:

Here we list all the factors limiting the profit in each quarter, as well as the marginal change in profit if we increase their current respective limits by 1. If the change is negative, the limit should be decreased to increase the profit. If the change is zero, it exists but is practically negligible due to rounding. A more detailed overview, along with the upper and lower feasible boundaries of the limits, is listed quarterly in tables 7–10.

1st quarter Jan–Mar (table 7):

- Supply of Beef (£0.6) and supply of Moneymaker (£0.3)
- Contract of Juice (£-0.1), Sauce (£-0.2) and Paste (£-0.1)
- Demand of Canned (£0.3)

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2nd quarter Apr-Jun (table 8):

- Supply of Moneymaker (£0.0)
- Demand of each product (see table)

3rd quarter Jul-Sep (table 9):

- Supply of Moneymaker (£0.2)
- Demand of each product (see table)
- Production limit (£0.3)
- Warehouse size (£0.8)

4th quarter Oct-Dec (table 10):

- Supply of Beef (£0.5) and supply of Moneymaker (£0.3)
- Contract of Sauce (£-0.1)
- Demand of Canned (£0.3)

We make the following conclusions:

- Supply of Moneymaker is limiting the total profit in every quarter.
- Demand of Canned is limiting the total profit in every quarter.

By the analysis of limiting factors of profit we find the conclusion is consistent with the analysis result from our stock analysis: Moneymaker is the cheapest raw material we can use for production and Canned is the product which is made the most, even though Paste has a higher selling price (see detailed in production and sales breakdowns in tables 17–20).

Stage 2: New Scenarios

The company feels that demand and overall product prices (including prices outlined in contracts) will increase by 20%, but that product cost will also increase by 5%. TomMac has 3 strategic arrangements that are currently offered:

1. Rent a new warehouse of capacity 5,000kg at a cost of £3,000 per year.
2. Sign a new contract which is specified in the table (reference appendix).
3. Increase production limit by 20% at a cost of £400 per year.

The objective is to combine these arrangements to reach the optimal strategic decision. There are in total 8 possible strategic decisions because, for each strategic decision, each of the aforementioned arrangements can be either included or excluded. Out of these 8 possible strategic decisions, 2 are infeasible and in both of them the new contract was signed, but the new warehouse wasn't rented (see table 11 in appendix). To satisfy both contracts it is essential to secure additional warehouse space. The optimal strategy is achieved when the new contract is signed, the new warehouse is rented and the production limit is **NOT** increased. The profit of optimal strategy is **£56,041.7**.

Additional Analysis and Further Research

During our analysis, we have noticed that the content of Juice does not stay constant throughout the year, as can be observed in figure 2 (precise numbers in table 12 in appendix).

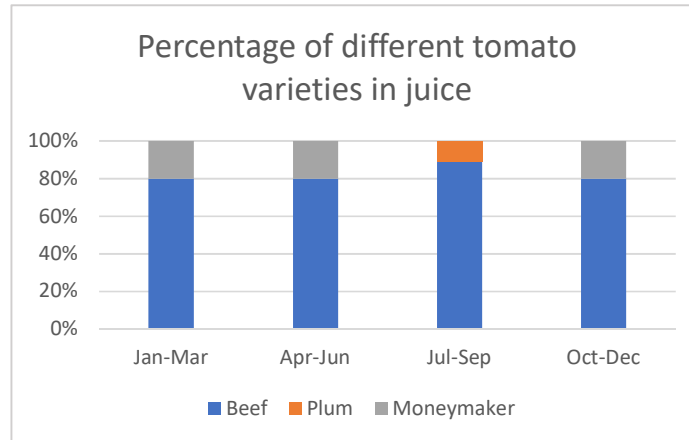


Figure 2. Percentage makeup of Juice per quarter.

Although the Juice produced in each quarter will have the same solid-liquid consistency, there is a chance that the product will have a different flavour or any other properties. We therefore recommend a feasibility investigation into creating two products: Juice and Juice Premium.

Additionally, as could be observed in figure 1 regarding storage, the number of products in stock decreases from Q3 to Q4. This means that Sauce and Paste produced in Q3 will make it to the market at the same time as the products produced in Q4. These products may, although having a constant solid-liquid ratio, may change in storage and therefore exhibit different properties not included in the model. We therefore recommend a quality analysis of the products in stock to make sure they are up to TomMac standards when they make the market.

Lastly, we propose a study of parameters which are currently not a part of the model: introducing emissions constraints, and exploring possibilities of introducing other new products, using other varieties of tomatoes as resources or expanding to new markets.

Conclusion and Recommendations

The model was able to capture the current company performance well, when allowing for negative profit. This means that current TomMac's operations are near optimal. Currently, the product with the highest selling price is Paste, but the product generating the most profit is Canned. Its production limit can be increased in all quarters to increase profit, as well as the supply of Moneymaker, the most versatile tomato variety.

Given current predictions, the most profitable future strategy is to sign the new available contract and to rent a new warehouse to accommodate it, but not to increase production. However, most future scenarios are profitable and may be chosen at TomMac's discretion. There is also room to refine the model and broaden current data insights.

Appendix

Quarter	Income(£)	Cost(£)	Profit(£)
Jan-Mar	28,095.0	19,755.2	8,339.8
Apr-Jun	37,505.0	30,478.1	7,026.9
Jul-Sep	37,505.0	41,440.6	-3,935.6
Oct-Dec	33,393.0	20,332.4	13,060.6
Year	136,498.0	112,006.3	24,491.7

Table 1. Annual profit breakdown.

Quarter	Income(£)	Cost(£)	Profit(£)
Jan-Mar	30,381.3	19,755.2	10,626.1
Apr-Jun	37,505.0	36,953.9	551.1
Jul-Sep	37,505.0	37,505.0	0.0
Oct-Dec	32,685.0	19,906.2	12,778.8
Year	138,076.3	114,120.3	23,956.0

Table 2. Annual profit breakdown when requiring non-negative quarterly profit.

Quarter	Contract(£)	Demand(£)	Total Income(£)
Jan-Mar	23,515.0	4,580.0	28,095.0
Apr-Jun	23,515.0	13,990.0	37,505.0
Jul-Sep	23,515.0	13,990.0	37,505.0
Oct-Dec	23,515.0	9,878.0	33,393.0
Year	94,060.0	42,438.0	136,498.0

Table 3. Annual income breakdown (aggregated for all products).

Quarter	Tomatoes(£)	Making Products(£)	Storing(£)	Total Cost(£)
Jan-Mar	8,733.3	11,021.9	0.0	19,755.2
Apr-Jun	9,967.6	20,501.5	9.0	30,478.1
Jul-Sep	9,785.1	31,155.5	500.0	41,440.6
Oct-Dec	8,959.5	11,198.8	174.1	20,332.4
Year	37,445.5	73,877.7	683.1	112,006.3

Table 4. Annual cost breakdown (aggregated for all products).

Quarter	Beef(kg)	Beef(£)	Plum(kg)	Plum(£)	Moneymaker(kg)	Moneymaker(£)
Jan-Mar	1,500.0	1,350.0	4,183.3	4,183.3	4,000.0	3,200.0
Apr-Jun	2,643.6	1,586.2	5,544.8	3,881.4	9,000.0	4,500.0
Jul-Sep	4,207.8	1,683.1	8,403.9	4,202.0	13,000.0	3,900.0
Oct-Dec	1,500.0	1,350.0	4,409.5	4,409.5	4,000.0	3,200.0
Year	9,851.4	5,969.3	22,541.6	16,676.2	30,000.0	14,800.0

Table 5. Total tomato varieties used per quarter (aggregated for all products).

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Quarter	Juice (kg)	Juice (£)	Sauce (kg)	Sauce (£)	Paste (kg)	Paste (£)	Canned (kg)	Canned (£)
Jan-Mar	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Apr-Jun	0.0	0.0	0.0	0.0	89.7	9.0	0.0	0.0
Jul-Sep	0.0	0.0	1,410.3	141.0	3,589.7	359.0	0.0	0.0
Oct-Dec	0.0	0.0	108.0	10.8	1,633.3	163.3	0.0	0.0
Year	0.0	0.0	1,518.3	151.8	5,312.7	531.3	0.0	0.0

Table 6. Breakdown of products in stock (quantity and storage cost).

Product Name	Limiting Factor	Unit Profit Change (£)	Range Start	Range End
Beef	Supply	0.6	957.1	1549.1
Moneymaker	Supply	0.3	2836.9	4336.9
Juice	Contract	-0.1	923.3	1848.2
Sauce	Contract	-0.2	1892	3000
Paste	Contract	-0.1	2275.4	3275.4
Canned	Demand	0.3	3652.5	6225

Table 7. Production constraints in quarter 1.

Product Name	Limiting Factor	Unit Profit Change (£)	Range Start	Range End
Moneymaker	Supply	0	8865.5	9542.3
Juice	Demand	0.6	1000	2840.9
Sauce	Demand	0.6	2204.7	3197.3
Paste	Demand	1	3138.5	3589.7
Canned	Demand	0.6	3000	5538.2

Table 8. Production constraints in quarter 2.

Product Name	Limiting Factor	Unit Profit Change (£)	Range Start	Range End
Moneymaker	Supply	0.2	11750	14757
Juice	Demand	0.7	1000	4000
Sauce	Demand	0.8	2000	5365.7
Paste	Production Limit	0.3	6786.3	7361.5
Paste	Demand	0.9	3138.5	3713.7
Canned	Demand	0.8	3000	10000
Warehouse Size	Warehouse Size	0.8	3805.7	5213.7

Table 9. Production constraints in quarter 3.

Product Name	Limiting Factor	Unit Profit Change (£)	Range Start	Range End
Beef	Supply	0.5	957.1	1597.1
Moneymaker	Supply	0.3	2836.9	4336.9
Sauce	Contract	-0.1	1786.3	3000
Canned	Demand	0.3	3652.5	5885.7

Table 10. Production constraints in quarter 4.

Strategy	Profit (£)	New Contract	New Warehouse	Incr. Production Lim.
1	52,445.7	✗	✗	✗
2	52,745.7	✗	✗	✓
3	53,978.0	✗	✓	✗
4	54,313.0	✗	✓	✓
5	Infeasible	✓	✗	✗
6	Infeasible	✓	✗	✓
7	56,041.7	✓	✓	✗
8	55,884.5	✓	✓	✓

Table 11. Feasibility and total profit of future strategies.

Quarter	Beef used	Plum used	Moneymaker used
Jan-Mar	80.0%	0.0%	20.0%
Apr-Jun	80.0%	0.0%	20.0%
Jul-Sep	88.9%	11.1%	0.0%
Oct-Dec	80.0%	0.0%	20.0%

Table 12. Percentage makeup of Juice per quarter.

Quarter	Beef used(kg)	Plum used(kg)	Moneymaker used(kg)
Jan-Mar	640.0	0.0	160.0
Apr-Jun	1,280.0	0.0	320.0
Jul-Sep	1,422.2	177.8	0.0
Oct-Dec	1,182.9	0.0	295.7

Table 13. Quantity of tomato varieties used for making Juice.

Quarter	Beef used(kg)	Plum used(kg)	Moneymaker used(kg)
Jan-Mar	860.0	0.0	1,290.0
Apr-Jun	1,363.6	0.0	2,045.5
Jul-Sep	2,785.6	976.1	1,250.0
Oct-Dec	317.1	0.0	475.7

Table 14. Quantity of tomato varieties used for making Sauce.

Quarter	Beef used(kg)	Plum used(kg)	Moneymaker used(kg)
Jan-Mar	0.0	433.3	1,300.0
Apr-Jun	0.0	1,794.8	5,384.5
Jul-Sep	0.0	3,500.0	10,500.0
Oct-Dec	0.0	659.5	1,978.6

Table 15. Quantity of tomato varieties used for making Paste.

Quarter	Beef used(kg)	Plum used(kg)	Moneymaker used(kg)
Jan-Mar	0.0	3,750.0	1,250.0
Apr-Jun	0.0	3,750.0	1,250.0
Jul-Sep	0.0	3,750.0	1,250.0
Oct-Dec	0.0	3,750.0	1,250.0

Table 16. Quantity of tomato varieties used for making Canned.

Quarters	Juice (kg)	Sauce (kg)	Paste (kg)	Canned (kg)
Jan-Mar	1,000.0	1,892.0	866.7	5,000
Apr-Jun	2,000.0	3,000.0	3,589.7	5,000
Jul-Sep	2,000.0	4,410.3	7,000.0	5,000
Oct-Dec	1,848.2	697.7	1,319.1	5,000
Year	6,848.2	10,000.0	12,775.4	20,000

Table 17. Quantities of products made per quarter.

Quarters	Juice Cost(£)	Sauce Cost(£)	Paste Cost(£)	Canned Cost(£)
Jan-Mar	1,000.0	2,855.2	2,166.7	5,000.0
Apr-Jun	2,000.0	4,527.3	8,974.2	5,000.0
Jul-Sep	2,000.0	6,655.5	17,500.0	5,000.0
Oct-Dec	1,848.2	1,052.9	3,297.6	5,000.0
Year	6,848.2	15,090.9	31,938.5	20,000.0

Table 18. Cost of making products per quarter.

Quarter	Juice Sold(kg)	Sauce Sold(kg)	Paste Sold(kg)	Canned Sold(kg)
Jan-Mar	1,000.0	2,000.0	2,500.0	5,000.0
Apr-Jun	2,000.0	3,000.0	3,500.0	5,000.0
Jul-Sep	2,000.0	3,000.0	3,500.0	5,000.0
Oct-Dec	1,848.2	2,000.0	3,275.4	5,000.0
Year	6,848.2	10,000.0	12,775.4	20,000.0

Table 19. Quantity of products sold per quarter.

Quarter	Juice(£)	Sauce(£)	Paste(£)	Canned(£)
Jan-Mar	3,890.0	7,610.0	15,215.0	10,790.0
Apr-Jun	3,890.0	7,610.0	15,215.0	10,790.0
Jul-Sep	3,890.0	7,610.0	15,215.0	10,790.0
Oct-Dec	3,890.0	7,610.0	15,215.0	10,790.0
Year	15,560.0	30,440.0	60,860.0	43,160.0

Table 20. Money earned from selling products per quarter (contract and demand).